

CHAPTER 18**ILOs**

By the end of the lectures, the student should be able to:

- *Name the muscles and ligaments attached to hyoid bone.*
- *Name the muscles and ligaments attached to styloid process.*
- *Name the cartilage membrane and ligaments of larynx.*
- *Name the ossicles, muscles, nerves and arteries in the middle ear.*
- *Mention the important anatomical events which occur at the level of cricoid cartilage C6 vertebra.*
- *Name the tonsils which form the Waldeyer's ring.*
- *Name the petrosal nerves and give their distribution.*
- *Name the parasympathetic ganglia in head and neck and give their sites.*

IMPORTANT SUMMARY OF HEAD AND NECK

Muscles and Ligaments Attached to Hyoid Bone
(Nine Muscles, Two Ligaments and One Membrane)

A- Muscles attached to hyoid bone:

- ***Two muscles originated from hyoid bone:***

Hyoglossus muscle (superficially) and middle constrictor muscle of pharynx (deeply) from the upper surface of the greater horn.

- ***Eight muscles are inserted into hyoid bone:***

Four suprahyoid muscles, 3 infrahyoid muscles and genioglossus muscle.

Suprahyoid muscles:

- Geniohyoid and mylohyoid muscles: Both are inserted into the anterior surface of body of hyoid.
- Stylohyoid muscle and fibrous loop of intermediate tendon of digastric muscle are attached to the junction of the body with the greater horn.

Genioglossus muscle is inserted in the anterior surface of the upper part of the body of the hyoid near the midline.

Infrahyoid muscles:

- Sternohyoid muscle (medially) and superior belly of omohyoid (laterally) both are inserted into the lower border of the body.
- Thyrohyoid muscle inserted into the lateral border of the greater horn.

B- Ligaments and membranes attached to hyoid bone:

- 1- **Stylohyoid ligament:** Attached to apex of the lesser horn.
- 2- **Hyoepiglottic ligament:** Attached to the upper border of the body
- 3- **Thyrohyoid membrane:** Attached to the upper part of the posterior surface of the body and to the medial border of the greater horn.

Styloid Apparatus

One bone (styloid process), 2 ligaments and 3 muscles.

Styloid muscles: Styloglossus (12th cranial nerve), stylohyoid (7th cranial nerve) and stylopharyngeus (9th cranial nerve).

Styloid ligaments: Stylomandibular and stylohyoid.

Nerve Supply of the Muscles of Head and Neck

- **All muscles of scalp and face** are supplied by **facial nerve**.
- All muscles of **mastication** are supplied by **mandibular nerve**.
- All **extraocular** muscles are supplied by the **oculomotor nerve** except:
 - a- **Superior oblique:** Supplied by **trochlear nerve**.
 - b- **Lateral rectus:** Supplied by **abducent nerve**.
- **Sternomastoid** and **trapezius** are supplied by the **spinal accessory nerve**.
- **Scalene muscles** are supplied by **lower 5 cervical nerves**.
- **All muscles of the back of the neck** are supplied by the **dorsal rami of the cervical nerves** except trapezius muscle. Muscles of the 4th layer (suboccipital muscles) are supplied by the dorsal ramus of C1 nerve.
- **All infrahyoid muscles** are supplied by **ansa cervicalis** except thyrohyoid which is supplied by fibers from C1, via hypoglossal nerve.
- **All muscles of the tongue** are supplied by **hypoglossal nerve** except palatoglossus which is supplied by cranial root of accessory nerve via pharyngeal branch of vagus (pharyngeal plexus).
- **All muscles of the soft palate** are supplied by **cranial root of accessory nerve** via the pharyngeal branch of vagus except tensor palati which is supplied by mandibular nerve.

- **All muscles of the pharynx** are supplied by **cranial root of accessory nerve** via pharyngeal branch of vagus except stylopharyngeus which is supplied by glossopharyngeal nerve.
- **All muscles of the larynx** are supplied by the **recurrent laryngeal nerve** except cricothyroid which is supplied by external laryngeal nerve.

Important Canals in the Skull

- 1- Carotid and facial canals and canal for tensor tympani (in the petrous temporal bone).
- 2- Optic, infraorbital and nasolacrimal canals in the orbit.
- 3- Incisive and greater palatine canals in the hard palate.
- 4- Pterygoid canal in the root of the pterygoid process.
- 5- Vomerovaginal and palatovaginal canals on either sides of the vomer.
- 6- Hypoglossal and condylar canals in the lateral part of occipital bone in relation to the occipital condyle.
- 7- Mandibular canal in the mandible.
- 8- Fine canal or canaliculus present in the following:
 - a- Anterior and posterior canaliculus for chorda tympani, present in the middle ear.
 - b- Mastoid canaliculus present in the lateral wall of the jugular fossa which transmits auricular branch of vagus.
 - c- Tympanic canaliculus present in the bony ridge between the carotid canal and the jugular foramen, transmitting tympanic branch of glossopharyngeal nerve.

Important Fissures in the Skull

- 1- Superior and inferior orbital fissures in the orbit.
- 2- Pterygomaxillary fissure between infratemporal and pterygopalatine fossae.
- 3- Squamotympanic fissure: In norma basalis externa between the squamous and tympanic parts of temporal bone. It is divided by a small bony ridge into petrosquamous fissure above and petrotympanic fissure below.

Important Fossae in the Skull

- 1- *In norma frontalis:*** Incisive and canine fossae in maxilla.
- 2- *In the orbit:*** Lacrimal fossa for lacrimal gland in the anterolateral part of the roof of the orbit.
- 3- *In norma lateralis:***
 - a- Temporal fossa: Above zygomatic arch.
 - b- Infratemporal fossa: Below zygomatic arch and deep to the ramus of mandible.
 - c- Pterygopalatine fossa: Below and behind the apex of the orbit. It contains the maxillary nerve, 3rd part of maxillary artery and the sphenopalatine ganglion.
- 4- *In norma basalis externa:***
 - a- Pterygoid and scaphoid fossae in pterygoid process.
 - b- Mandibular fossa: Below and lateral to the root of the zygomatic arch.
 - c- Digastric fossa: On the medial surface of the mastoid process.
 - d- Jugular fossa: Deep to jugular foramen.
 - e- Condylar fossa: Behind the occipital condyle.
- 5- *In norma basalis interna (cranial cavity):***
 - a- Anterior, middle and posterior cranial fossae.
 - b- Pituitary fossa on the upper surface of the body of sphenoid.
- 6- *In mandible:*** Submandibular and sublingual fossae for the corresponding glands and digastric fossa for anterior belly of digastric muscle.

Cartilages of Larynx

- Three single (thyroid, cricoid and epiglottis).
- Three paired (arytenoid, corniculate and cuneiform).

Important Anatomical Events at the Level of the Cricoid Cartilage (C6)

- 1- Termination of pharynx and the beginning of oesophagus.
- 2- Termination of the larynx and the beginning of trachea.
- 3- The vertebral artery enters the foramen transversarium of 6th cervical vertebra.
- 4- Middle cervical sympathetic ganglion.
- 5- The upper convexity of inferior thyroid artery.
- 6- The crossing of intermediate tendon of omohyoid to the carotid sheath.
- 7- Middle thyroid vein emerges from thyroid gland.

Contents of Middle Ear

Air, 2 muscles (stapedius and tensor tympani), 3 ossicles (malleus, incus and stapes), 3 nerves (facial, chorda tympani and tympanic plexus) and many arteries (anterior tympanic from maxillary, posterior tympanic from stylomastoid, superior tympanic from middle meningeal and caroticotympanic from ICA and inferior tympanic from ascending pharyngeal artery).

Tonsils (Adenoid Tissue) (Waldeyer's Ring)

- 1- Nasopharyngeal tonsil lies at the junction of roof and posterior wall of the nasopharynx.
- 2- Tubal tonsil lies around the opening of the auditory tube in nasopharynx.
- 3- Palatine tonsil lies in the lateral wall of the oropharynx.
- 4- Lingual tonsil lies on the dorsum of the pharyngeal part of the tongue.

Arterial Supply of Palatine Tonsil

- 1- Tonsillar branch and ascending palatine artery (from facial).
- 2- Ascending pharyngeal artery from ECA.
- 3- Dorsal lingual arteries from 2nd part of lingual arteries.

Summary of the Distribution of Cranial Nerves

Sensory cranial nerves (1, 2, 8):

- **1st cranial nerve (olfactory nerve):** For the sense of smell.
- **2nd cranial nerve (optic nerve):** For vision.
- **8th cranial nerve (vestibulocochlear nerve):** For hearing (cochlear nerve) and equilibrium (vestibular nerve).

Motor cranial nerves (3, 4, 6, 11, 12):

- **3rd cranial nerve (oculomotor nerve):** Supplies all extraocular muscles except the lateral rectus and the superior oblique muscles. It also supplies two intraocular muscles: sphincter pupillae and ciliary muscles.
- **4th cranial nerve (trochlear nerve):** Supplies the superior oblique muscle.
- **6th cranial nerve (abducent nerve):** Supplies the lateral rectus muscle.
- **11th cranial nerve (accessory nerve):** It has 2 roots:
 - a- Spinal root supplies sternomastoid and trapezius muscles.
 - b- Cranial root supplies all muscles of soft palate except tensor palati and all muscles of pharynx except stylopharyngeus.
- **12th cranial nerve (hypoglossal nerve):** Supplies all muscles of tongue except palatoglossus.

Mixed cranial nerves (5, 7, 9, 10):

- **5th cranial nerve (trigeminal nerve):** It has 2 roots
 - a- **Motor root:** It joins and distributes with the mandibular nerve to supply 8 muscles: 4 muscles of mastication, 2 tensor muscles: tensor palati and tensor tympani and 2 other muscles mylohyoid and anterior belly of digastric.
 - b- **Sensory root:** It supplies the skin of the scalp in front of the auricle, skin of the face except the part covering the angle of mandible and parotid gland. It also supplies the general sensation of the anterior 2/3 of tongue.

Cutaneous Branches of Trigeminal Nerve

(A) Ophthalmic nerve	(B) Maxillary nerve	(C) Mandibular nerve
1. Supra-trochlear from frontal 2. Supra-orbital nerve 3. Infra-trochlear from nasociliary 4. External nasal nerve 5. Palpebral br. of lacrimal nerve	1. Zygomaticotemporal 2. Zygomaticofacial 3. Infraorbital gives 3 branches: palpebral, nasal, and labial	1. Auriculotemporal nerve 2. Buccal nerve 3. Mental nerve

- **7th cranial nerve (facial nerve):** It has 2 parts:
 - a- Motor part: supplies all muscles of the face, scalp, posterior belly of digastric, stylohyoid, stapedius and platysma.
 - b- Nervus intermedius (parasympathetic and sensory parts):
 - 1- Parasympathetic fibers supply submandibular, sublingual and lacrimal glands and glands of nose, palate and pharynx.
 - 2- Sensory fibers carry the taste sensation from anterior 2/3 of tongue.
- **9th cranial nerve (glossopharyngeal nerve):** It has 3 types of fibers:
 - 1- Motor: Supplies the stylopharyngeus muscle.
 - 2- Parasympathetic: Supplies the parotid gland.
 - 3- Sensory: for the soft palate, palatine tonsil, oropharynx, middle ear and posterior 1/3 of tongue (general sensation and taste), carotid sinus and carotid body.
- **10th cranial nerve (vagus nerve):** It has 3 types of fibers:
 - 1- Motor fibers supply all muscles of larynx, muscles of palate except tensor palati and muscles of pharynx except stylopharyngeus.
 - 2- Parasympathetic fibers supply the smooth muscles of GIT, trachea, bronchi, bronchioles and heart.
 - 3- Sensory fibers for the larynx, laryngopharynx, external auditory meatus and external surface of the ear drum.

Nerve Supply of Larynx

Motor: RLN for all muscles of larynx except cricothyroid by external laryngeal nerve.

Sensory: RLN supplies the mucosa below the vocal cords.

Internal LN supplies the mucosa above the vocal cords.

Nerve Supply of Pharynx

Motor: All muscles of pharynx except stylopharyngeus are supplied by pharyngeal plexus (cranial root of accessory).

Sensory:

- 1- Pharyngeal branches of the sphenopalatine ganglion (maxillary nerve) for nasopharynx.
- 2- Glossopharyngeal nerve via the pharyngeal plexus for oropharynx.
- 3- Internal laryngeal nerve for laryngopharynx.

Parasympathetic Ganglia in Head and Neck

- 1- **Ciliary ganglion** in the apex of the orbit, suspended from nasociliary nerve (of ophthalmic).
- 2- **Sphenopalatine ganglion** lies in the pterygopalatine fossa suspended from the maxillary nerve.
- 3- **Otic ganglion** lies in infratemporal fossa below foramen ovale suspended from the trunk of mandibular nerve.
- 4- **Submandibular ganglion** lies in the submandibular region on lateral surface of hyoglossus muscle suspended from the lingual nerve.

Petrosal Nerves

- 1- **Greater superficial petrosal nerve:** It is parasympathetic nerve which arises from facial nerve and relays in the sphenopalatine ganglion. Postganglionic fibers supply the lacrimal, pharyngeal, nasal and palatine glands.
- 2- **Lesser superficial petrosal nerve:** It is parasympathetic nerve which arises from the tympanic plexus and relays in the otic ganglion. Postganglionic fibers supply the parotid gland.
- 3- **Deep petrosal nerve:** It is sympathetic nerve which arises from the sympathetic plexus around ICA. It traverses the sphenopalatine ganglion without relay to supply the glands of nose, palate, pharynx and lacrimal glands.